Flow Control in Data Link Layer

1. Basic Stop-and-Wait Protocol

- The Stop-and-Wait Protocol is a simple data communication technique in which the **sender transmits only one frame at a time and then waits until an acknowledgment (ACK) is received before sending the next frame.**
- It is mainly used for flow control, ensuring that the receiver is not overloaded.

Working of Stop-and-Wait Protocol

1. The sender sends a single data frame to the receiver.
2. After sending, the sender stops and waits for an acknowledgment.
3. The receiver receives the frame successfully.
4. The receiver sends an acknowledgment (ACK) back to the sender.
5. Only after receiving the ACK, the sender transmits the next frame.

Key Characteristics

- Only one frame is sent at a time.
- Sender must wait for ACK before sending the next frame.
- Very easy to implement.

Limitations of Basic Stop-and-Wait

1. **Low Efficiency**
   - The sender remains idle while waiting for ACK.
   - This wastes bandwidth, especially in long-distance networks.
2. **No Error Control**
   - If a frame is lost or corrupted, the protocol does not handle retransmission.
3. **Not Suitable for Noisy Channels**
   - Works well only in error-free environments.

Stop-and-Wait ARQ Protocol

- Stop-and-Wait ARQ (Automatic Repeat request) is an **improved version** of Stop-and-Wait that provides both: **Flow Control and Error Control**
- In this protocol, the sender retransmits the frame automatically if:
  - the frame is lost or damaged
  - the acknowledgement is not received within a certain time
- It is used in connection-oriented communication, where a logical connection is established before data transfer.
- It uses only two sequence numbers (0 and 1) to identify frames and avoid duplication.
- The throughput is one data frame per round-trip time (RTT), which leads to poor performance when the bandwidth–delay product is high.

Why ARQ is Needed?

In real communication channels, errors may occur such as:

- Frame loss
- Frame corruption
- ACK loss

To ensure reliable transmission, ARQ introduces:

- Timer
- Retransmission
- Sequence numbers

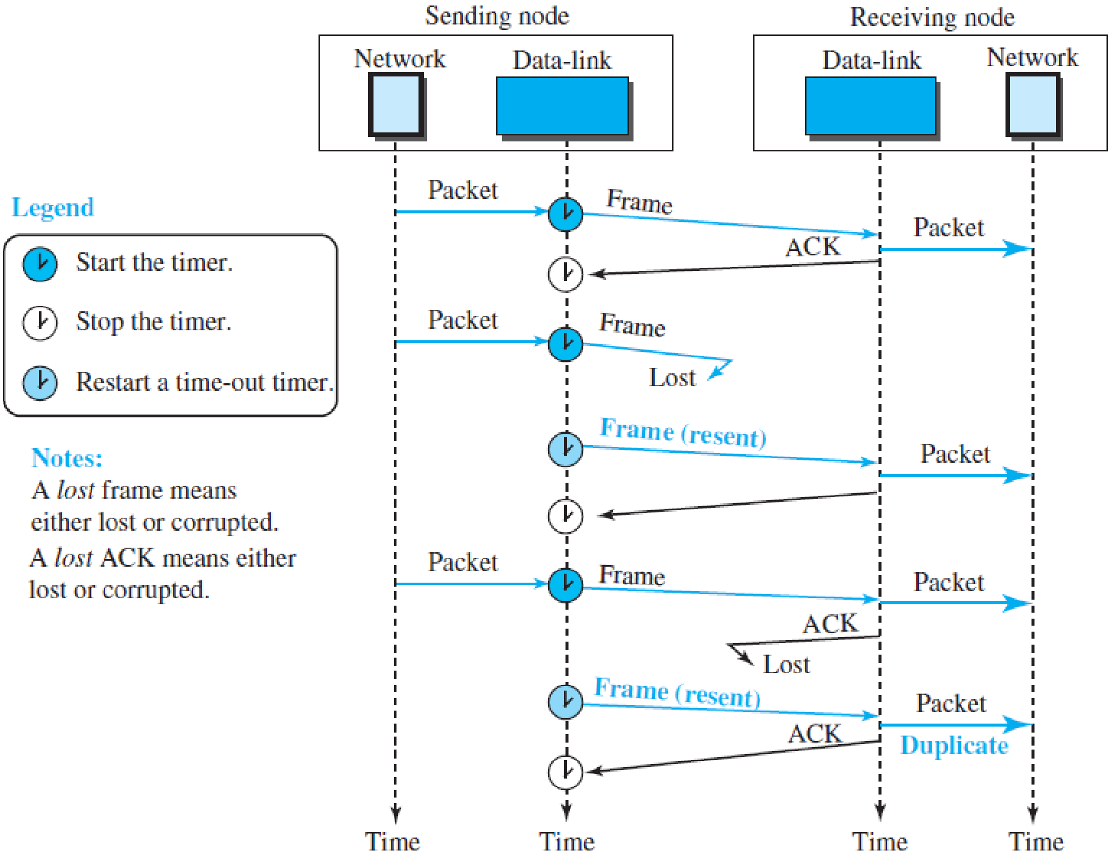
Useful Terms in the Stop and Wait Protocol

- **Propagation Delay:** The time taken by a data packet to travel physically from the sender to the receiver over the communication medium.
- **Roundtrip Time (RTT):** The total time taken for a data frame to reach the receiver plus the time taken for the acknowledgment (ACK) to return to the sender.

- **Timeout (TO):** The maximum time the sender waits for an acknowledgment before retransmitting the frame. Timeout = 2×RT
- **Time To Live (TTL):** The maximum time a frame is allowed to exist in the network before being discarded to prevent infinite looping. TTL = 2×Timeout.

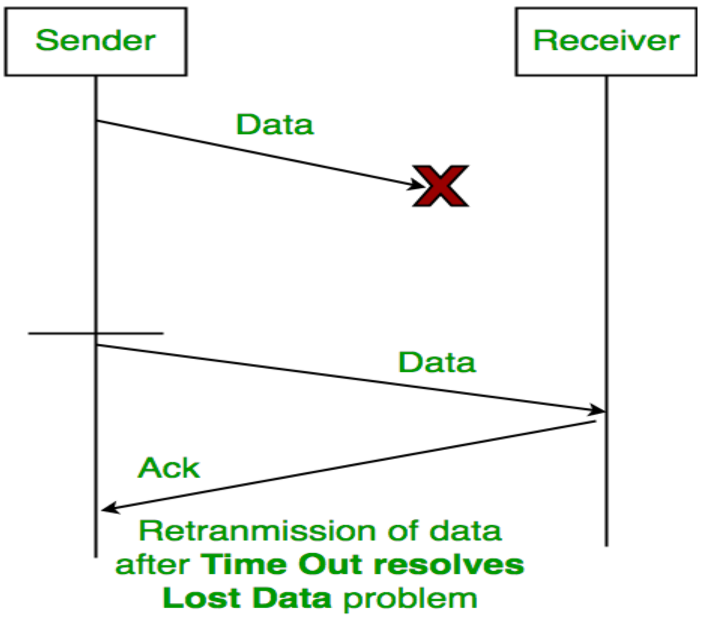
Working of Stop-and-Wait ARQ

1. Sender sends one frame (Frame 0).
2. Sender starts a timer.
3. Receiver checks the frame:
    - If correct, then sends ACK
    - If corrupted, then discards frame (no ACK)
4. Sender waits for ACK:
    - If ACK received , then sends next frame
    - If ACK not received and timeout occurs, then frame is resent

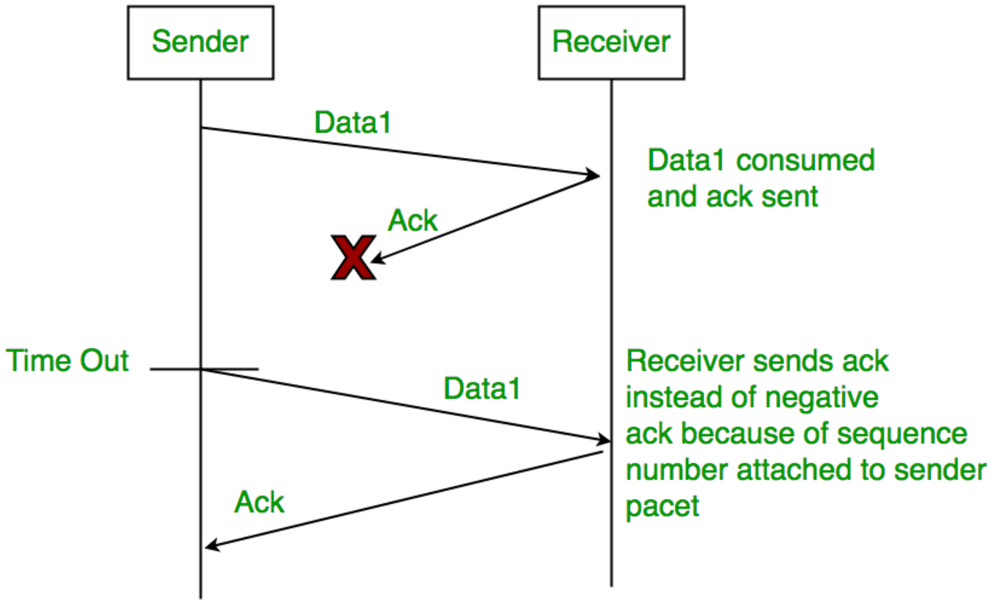


1. TimeOut

Case 1: Frame Lost



Case 2: ACK Lost



2. Sequence Number (Data) -

- In Stop-and-Wait ARQ, **each data frame is assigned a sequence number by the sender.**
- This helps the receiver uniquely identify each frame, detect duplicate frames, and send the correct acknowledgment.
- The sender transmits the next frame only after receiving the acknowledgment for the current frame, ensuring reliable data delivery.

3. Sequence Number

- In Stop-and-Wait ARQ, acknowledgments (ACKs) also carry sequence numbers.
- After successfully receiving a data frame, the receiver sends an ACK indicating the sequence number of the next expected frame.
- The sender uses this information to confirm successful delivery and decide whether it can proceed with the next transmission or needs to retransmit a frame.

Stop-and-Wait ARQ: Formulas

1. **Frame Transmission Time:**

Time taken to transmit one full frame.

If:

- Frame size = L bits
- Bandwidth = R bps

$$T_t = \frac{L}{R}$$

2. **Propagation Delay**

Time taken for signal to travel one way.

If:

- Distance = d meters
- Propagation speed = v m/s

$$T_p = \frac{d}{v}$$

3. **Round Trip Time (RTT)**

$$RTT = 2T_p$$

4. **Total Cycle Time (One Frame + ACK)**

One complete cycle:

- Send frame + Wait for ACK

$$T_{cycle} = T_t + 2T_p$$

(If ACK time included)

$$T_{cycle} = T_t + 2T_p + T_{ack}$$

5. **Throughput**

Useful data delivered per unit time:

$$Throughput = \frac{L}{T_t + 2T_p}$$

6. **Efficiency (Channel Utilization)**

Fraction of time channel is actually transmitting:

$$Efficiency = \frac{T_t}{T_t + 2T_p}$$

Efficiency in Terms of "a"

Define:

$$a = \frac{T_p}{T_t}$$

Then:

$$Efficiency = \frac{1}{1 + 2a}$$

Relation Between Throughput and Efficiency

$$Throughput = Efficiency \times Bandwidth$$

Stop-and-Wait ARQ Numericals

Given:

- Frame size $L = 2000$bits
- Bandwidth $R = 1$Mbps
- Propagation delay $T_p = 10$ms

Find:

1. Transmission time
2. Efficiency
3. Throughput

Step 1: Transmission Time

$$T_t = \frac{L}{R}$$

$$T_t = \frac{2000}{10^6} = 0.002s = 2ms$$

Step 2: Cycle Time

$$T_{cycle} = T_t + 2T_p$$

$$T_{cycle} = 2ms + 2(10ms)$$

$$T_{cycle} = 2ms + 20ms = 22ms$$

Step 3: Efficiency

$$Efficiency = \frac{T_t}{T_t + 2T_p}$$

$$Efficiency = \frac{2}{22} = 0.0909$$

$$Efficiency \approx 9.1\%$$

Step 4: Throughput

$$Throughput = Efficiency \times Bandwidth$$

$$Throughput = 0.0909 \times 1Mbps$$
$$Throughput \approx 0.0909Mbps = 90.9kbps$$

- Efficiency = **9.1%**
- Throughput = **90.9 kbps**

Practice Questions on Stop and Wait

Question 1

Given:

- Frame size $L = 3000$bits
- Bandwidth $R = 1.5$Mbps
- Propagation delay $T_p = 12$ms

Find:

1. Transmission time
2. Efficiency
3. Throughput

Question 2

Given:

- Frame size $L = 5000$bits
- Bandwidth $R = 1$Mbps
- Distance = 2000 km
- Propagation speed $v = 2.5 \times 10^8$m/s

Find:

1. Propagation delay
2. Efficiency
3. Throughput

Sliding Window Protocol

Sliding Window Protocol is a method of data transmission in which:

- The sender can send multiple frames before waiting for acknowledgment.
- The receiver can accept multiple frames depending on its capacity.

A "window" is maintained at both sender and receiver side.

This window represents the range of frames that can be transmitted or received at a given time.

Need of Sliding Window Protocol

Stop-and-Wait protocol has several drawbacks:

Low Channel Utilization

- The sender remains idle while waiting for acknowledgment.
- This wastes bandwidth.

Poor Throughput

- Only one frame is in transit at a time.
- This reduces the overall data transfer rate.

Not Suitable for Long Distance Communication

- If propagation delay is high, the waiting time becomes very large.

Sliding Window Protocol solves these problems by allowing continuous transmission.

Working of Sliding Window Protocol

Sliding Window Protocol uses a technique called pipelining.

1. Sender has a window size of W frames.
2. Sender can transmit all frames within the window without waiting for acknowledgments.
3. Receiver receives frames and sends acknowledgments.
4. When sender receives acknowledgment for the earliest frame, the window moves forward.
5. New frames are added into the window and transmission continues.

This moving window mechanism is called "sliding".

Example

If sender window size is 4, sender can send frames:

Frame 0
Frame 1
Frame 2
Frame 3

After receiving acknowledgment for Frame 0, sender window slides forward and sender can send Frame 4.

Now window becomes:

Frame 1
Frame 2
Frame 3
Frame 4

Important Concepts in Sliding Window

1. **Window Size**

- Window size refers to the maximum number of frames that can be sent without waiting for acknowledgment.
- Larger window size improves throughput.

2. **Sender Window**

Sender window contains frames that are:

- Ready to be sent
- Already sent but not yet acknowledged

3. **Receiver Window**

Receiver window contains frames that receiver is ready to accept.

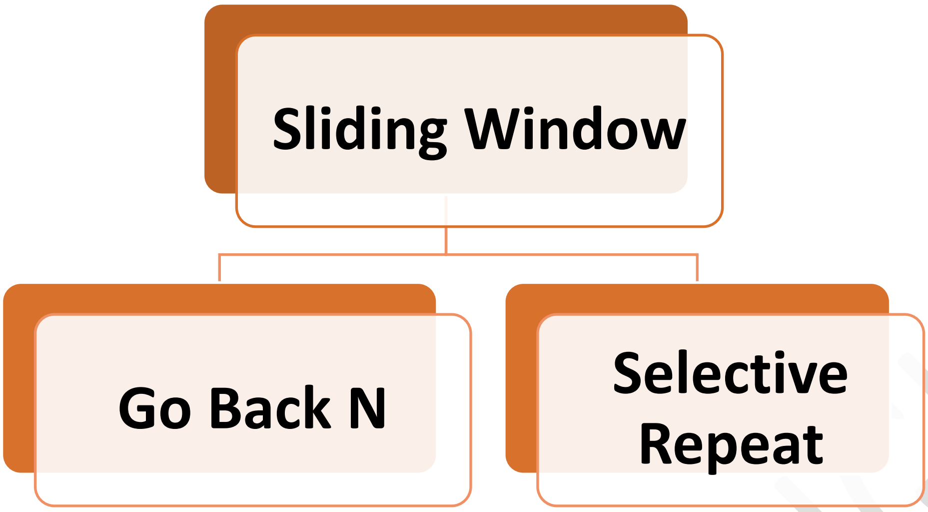
4. **Sequence Numbers**

Each frame is assigned a sequence number so that sender and receiver can identify frames correctly.

Sequence numbers help in:

- Detecting missing frames
- Avoiding duplicate frames
- Maintaining proper order

TYPES OF SLIDING WINDOW



Go-Back-N ARQ

Go-Back-N ARQ is an error/flow control protocol in which:

- Sender can send multiple frames.
- Receiver accepts only frames in correct order.
- If a frame is lost or damaged, all subsequent frames are discarded.
- Sender retransmits from the lost frame onward.

Working of Go Back N

1. Sender sends frames continuously.
2. Receiver sends **cumulative acknowledgments**.
3. If a **frame is missing, receiver cannot accept further frames**.
4. Sender **retransmits all frames** starting from the missing one.

Example Scenario

Sender sends frames:

0, 1, 2, 3

As ACK0 received, Frame 4 was sent. As ACK1 received, Frame 5 was sent.

Frame 2 is lost.

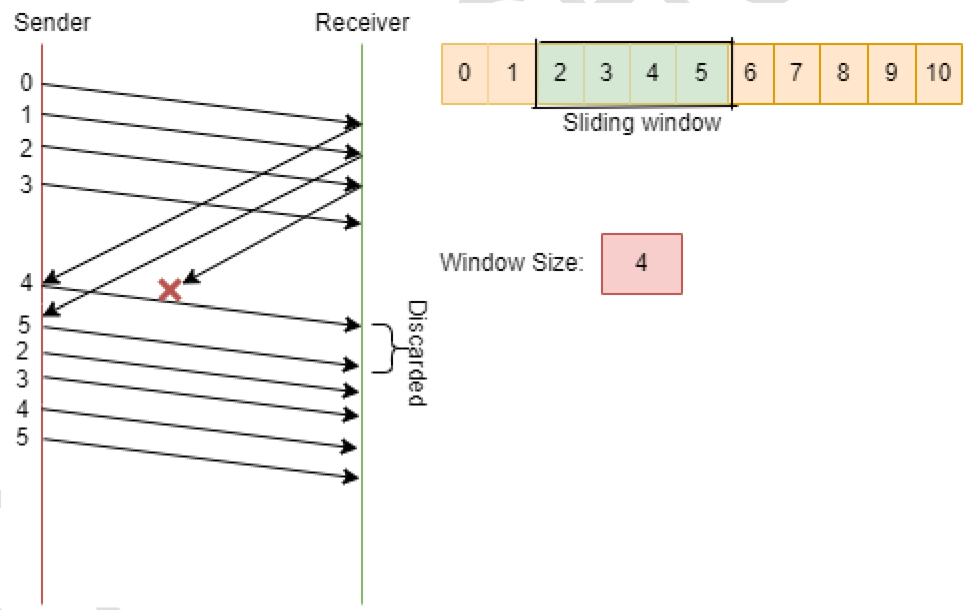
Receiver receives:

0, 1, then missing frame, then frames 3 arrive.

Receiver discards frame 3

Sender retransmits:

2, 3, 4,5



If any packet is lost, the receiver sends a negative acknowledgement (NACK) for the lost packet, and the sender retransmits all the packets in the window starting from the lost packet. The sender also maintains a timer for each packet, and if an acknowledgement is not received within the timer's timeout period, the sender retransmits all packets in the window.

What is Cumulative ACK?

The receiver acknowledges not only the last received frame, but also confirms that all previous frames have been received correctly in sequence.

So, instead of sending separate acknowledgements for every single frame again and again, the receiver sends an ACK that represents the highest correctly received frame in order.

Example:

If the receiver sends:

$$ACK\ k$$

It means:

- All frames from 0 up to k have been received successfully.
- The receiver is now expecting frame: $\boldsymbol{k+1}$

Example:

If receiver sends: $ACK2$
It indicates:

Frame 0 received
Frame 1 received
Frame 2 received

Now receiver expects frame 3.

Case 1: Normal Working of Cumulative ACK (No Loss Case)

Suppose sender transmits frames:

0,1,2,3

Receiver receives them correctly.

Then receiver sends acknowledgements:

ACK0,ACK1,ACK2,ACK3

Each ACK confirms all frames up to that number.

Case 2: Cumulative ACK in Case of Frame Loss

The most important use of cumulative ACK is seen when a frame is lost.

Example Scenario:

Sender sends frames:

$$0,1,2,3$$

Assume frame 2 is lost during transmission.

Receiver receives:

- Frame 0
- Frame 1
- Frame 2 : missing
- Frame 3 arrives but is discarded
- Frame 4 arrives but is discarded

Receiver,

In Go-Back-N, the **receiver accepts only frames in correct order.**

Since frame 2 is missing, receiver cannot accept frame 3 or frame 4.

So receiver keeps sending:

$$\boldsymbol{ACK1}$$

ACK Pattern in This Case:

$$\boldsymbol{ACK0, ACK1, ACK1} \ldots$$

Receiver repeats the same ACK until the missing frame arrives.

Cumulative ACK is the default acknowledgement method in Go-Back-N; it becomes more noticeable when a frame is lost because the receiver keeps repeating the last in-order ACK.

Key features of the Go-Back-N (GBN) protocol

- Sliding window mechanism
- Sequence numbers
- Cumulative acknowledgements
- Timeout mechanism
- NACK mechanism
- Simple implementation.

Sequence Numbers in Go-Back-N

Each transmitted frame is assigned a **sequence number**.

If the sequence number field contains **m bits**, then the **total number of sequence numbers available is: 2^m**

Example:

If $m = 3$,then $2^3 = 8$
Sequence numbers will be:

$$0,1,2,3,4,5,6,7$$

Since the sequence number range is limited, after reaching the maximum value, numbering starts again from 0.

This is called **wrap-around** or **sequence number reuse**:

$$0,1,2,\ldots,7,0,1,2,\ldots$$

Window Mechanism in Go-Back-N

Go-Back-N uses a sliding window at the sender side:

- The sender can send several frames within the window without waiting.
- The receiver window size is always **1**, meaning it only accepts the next expected frame.

Window Size Formula in Go-Back-N

The maximum sender window size in Go-Back-N is:

$$\boldsymbol{W \leq 2^m - 1}$$

Where:

- W= sender window size
- m= number of bits in the sequence number field

Go-Back-N ARQ Efficiency

Go-Back-N follows sliding window, but if one frame is lost, multiple frames are retransmitted.

Basic efficiency (without errors):

$$Efficiency = (W \times Tt)/(Tt + 2Tp)$$

With errors, efficiency reduces due to retransmissions.

Total Time in One Transmission Cycle

In Go-Back-N, sender can transmit W frames back-to-back.

So total transmission time for W frames: $\mathbf{W \times T_t}$

But sender must still wait for acknowledgment to return.

Round Trip propagation delay: $\mathbf{2T_p}$

So total cycle time becomes:

Cycle Time=W * T_t + $2T_p$

Efficiency of Go-Back-N

Efficiency means:

- How much time the channel is actually sending useful data.

So efficiency is:

$$\boldsymbol{Efficiency = (W \times Tt)/(W \times Tt + 2Tp)}$$

$$\mathbf{= W/1+2*a \ (Here\ a = T_p/T_t)}$$

Special Case: Full Utilization

If window size is very large such that:

$$W \times Tt \geq 2Tp$$

Then channel never becomes idle.

So:

$$Efficiency = 1$$

That means 100 percent utilization.

Throughput of Go-Back-N

Throughput means:

- Actual useful data delivered per second.

Throughput is:

$$Throughput = Efficiency \times Bandwidth$$

So:

$$Throughput = ((W \times Tt)/(W \times Tt + 2Tp)) \times R$$

Numerical Questions

Numerical 1

Given:

- Frame size $L = 2000\text{bits}$
- Bandwidth $R = 1\text{Mbps}$
- Propagation delay $Tp = 10\text{ms}$
- Window size $W = 4$

Find:

1. Transmission time
2. Efficiency
3. Throughput
4. Minimum number of sequence number bits required

Solution:

Transmission Time

$$Tt = L/R$$
$$Tt = 2000/10^6$$
$$Tt = 0.002s = 2ms$$

Total Useful Transmission Time for W frames

$$W \times Tt = 4 \times 2ms = 8ms$$

Cycle Time

$Cycle\ Time = (W \times Tt) + 2Tp$
$Cycle\ Time = 8ms + 2(10ms)$
$Cycle\ Time = 8ms + 20ms = 28ms$

Efficiency

$$Efficiency = (W \times Tt)/(W \times Tt + 2Tp)$$

$$Efficiency = 8/28$$
$$Efficiency = 0.2857$$
$$Efficiency \approx 28.6\%$$

Throughput

$$Throughput = Efficiency \times Bandwidth$$
$$Throughput = 0.2857 \times 1Mbps$$
$$Throughput = 0.2857Mbps = 285.7kbps$$

For **Go-Back-N protocol**:

$$W \leq 2^k - 1$$
$$4 \leq 2^k - 1$$
$$2^k \geq 5$$

Smallest value satisfying this:

$$k = 3$$

Therefore, the Minimum sequence number bits required = 3 bits

Practice Question 1: Go back N

Given:

- Frame size $L = 3000$bits
- Bandwidth $R = 1.5$Mbps
- Propagation delay $Tp = 12$ms
- Window size $W = 4$

Find:

1. Transmission time
2. Efficiency
3. Throughput
4. Minimum number of sequence number bits required

Selective Repeat ARQ

Selective Repeat ARQ is a type of Sliding Window Protocol used in the Data Link Layer.

It is designed to provide:

- Reliable data transmission
- Better efficiency than Go-Back-N
- Improved performance when errors occur

Selective Repeat is an advanced error control mechanism.

Selective Repeat ARQ is an Automatic Repeat Request protocol in which:

- The sender can transmit multiple frames continuously.
- The **receiver can accept frames even if they arrive out of order.**
- Only the **lost or damaged frames are retransmitted.**

Unlike Go-Back-N, it does not retransmit all frames after an error.

Why is Selective Repeat Needed?

Go-Back-N ARQ has one major limitation:

- If one frame is lost, all following frames are retransmitted even if they were received correctly.

This leads to:

- Bandwidth wastage
- Lower throughput
- Poor performance in noisy channels

Selective Repeat solves this problem by retransmitting only the missing frame.

Working of Selective Repeat ARQ

Selective Repeat uses sliding windows at both sender and receiver side.

Step 1: Sender Window Operation

- Sender has a window size W.
- Sender can send W frames without waiting for acknowledgment.

Example:

If window size is 4, sender can send:

Frame 0
Frame 1
Frame 2
Frame 3

Step 2: Receiver Window Operation

- Receiver also maintains a window.
- Receiver accepts frames even if they come out of order.
- Receiver stores the frames in a buffer until missing frames arrive.

Step 3: Individual Acknowledgments

In Selective Repeat:

- Receiver sends separate acknowledgment for each correctly received frame.

This is different from Go-Back-N where cumulative acknowledgment is used.

Step 4: Handling Frame Loss

If a frame is lost during transmission:

- Receiver continues to accept other frames.
- Receiver stores them in buffer.

- Sender retransmits only the missing frame.

Example

Sender sends frames:

Frame 0, Frame 1, Frame 2, Frame 3, Frame 4

Suppose Frame 2 is lost.

Receiver receives:

Frame 0 (accepted)
Frame 1 (accepted)
Frame 3 (accepted and stored)
Frame 4 (accepted and stored)

Receiver sends acknowledgments for:

Frame 0
Frame 1
Frame 3
Frame 4

Sender notices Frame 2 acknowledgment is missing.

Sender retransmits only:

Frame 2

After Frame 2 arrives, the receiver delivers all frames in the correct order.

Window Size in Selective Repeat ARQ

Selective Repeat ARQ allows multiple frames to be sent before receiving acknowledgments.Both sender and receiver maintain a **sliding window**.

However, window size must be limited to avoid confusion between:

- New frames
- Old delayed duplicate frames

Maximum Window Size Formula

In Selective Repeat ARQ, the window size must satisfy:

$$W \leq 2^{(m-1)}$$

Where:

- W= window size
- m= number of sequence number bits

Why Window Size is Limited?

If window size is too large, the receiver may not be able to distinguish whether a frame is:

- A new frame OR A duplicate frame from an earlier transmission

Therefore, the window size is restricted to **half of the sequence number space**.

Protocol	Max Window Size
Selective Repeat	(2^{m-1})
Go-Back-N	$(2^m - 1)$

Key Features of Selective Repeat ARQ

Only Lost Frames are Retransmitted

- Selective Repeat retransmits only the missing or corrupted frame.
- This reduces unnecessary retransmissions.

Receiver Accepts Out-of-Order Frames

- Receiver does not discard correctly received frames.
- It stores them until the missing frame arrives.

Buffering is Required

- Receiver must have enough memory to store out-of-order frames.

Efficiency and Throughput in Selective Repeat

Selective Repeat uses the sliding window efficiency formula.

Let:

- Transmission time of one frame = T_t
- Propagation delay = Tp
- Window size = W

Efficiency

$$\boldsymbol{Efficiency = (W \times Tt)/(W \times Tt + 2Tp)}$$

Throughput

$$Throughput = Efficiency \times Bandwidth$$

Selective Repeat gives higher throughput in error-prone networks because fewer frames are retransmitted.

Advantages of Selective Repeat ARQ

- Higher efficiency than Go-Back-N
- Better bandwidth utilization
- Only lost frames are retransmitted
- Suitable for noisy communication channels

Disadvantages of Selective Repeat ARQ

- Receiver needs buffering memory
- More complex implementation
- Requires individual acknowledgments
- More processing overhead

Selective Repeat vs Go-Back-N

Feature	Go-Back-N	Selective Repeat
Receiver accepts frames	Only in order	In order and out of order
Retransmission	Many frames	Only missing frames
Buffer requirement	Low	High
Complexity	Simple	Complex
Efficiency in noisy channel	Lower	Higher
Window Size	$W \leq 2^m - 1$	$W \leq 2^{m-1}$

Numerical on Sliding Window

Given:

- Frame size $L = 4000\text{bits}$
- Bandwidth $R = 2\text{Mbps}$
- Propagation delay $Tp = 15\text{ms}$
- Window size $W = 5$

Find:

1. Transmission time
2. Efficiency
3. Throughput

Step 1: Transmission Time

$$Tt = L/R$$
$$Tt = 4000/(2 \times 10^6)$$
$$Tt = 0.002s = 2ms$$

Step 2: Useful Transmission Time for W frames

$$W \times Tt = 5 \times 2ms = 10ms$$

Step 3: Cycle Time

$$Cycle\ Time = (W \times Tt) + 2Tp$$
$$Cycle\ Time = 10ms + 2(15ms)$$
$$Cycle\ Time = 10ms + 30ms = 40ms$$

Step 4: Efficiency

$$Efficiency = (W \times Tt)/(W \times Tt + 2Tp)$$
$$Efficiency = 10/40$$
$$Efficiency = 0.25$$
$$Efficiency = 25\%$$

Step 5: Throughput

$$Throughput = Efficiency \times Bandwidth$$
$$Throughput = 0.25 \times 2Mbps$$
$$Throughput = 0.5Mbps = 500kbps$$

Practice Question 1

Given:

- Frame size $L = 3000$bits
- Bandwidth $R = 1.5$Mbps
- Propagation delay $Tp = 12$ms
- Window size $W = 4$

Find:

1. Transmission time
2. Efficiency
3. Throughput
4. Minimum no. of Sequence Number Bits Required

PRACTICE QUESTIONS ON ARQ PROTOCOLS

Q1. A Go-Back-N ARQ protocol uses sequence numbers with **m = 5 bits**. Find the maximum possible sender window size.

Q2. A Selective Repeat ARQ system uses sequence numbers with **m = 4 bits**. Find the maximum sender window size.

Q3. A Go-Back-N ARQ system has window size = 7, frame transmission time = 3 ms, and RTT = 21 ms. Calculate the efficiency of the protocol.

Q4. A Stop-and-Wait ARQ system has frame transmission time = 4 ms and RTT = 20 ms. Find the efficiency of Stop-and-Wait ARQ.

Q5. A sender transmits 10 frames and frame number 3 is lost.

1. How many frames will be retransmitted in Go-Back-N ARQ?
2. How many frames will be retransmitted in Selective Repeat ARQ?